

Fuel Economy Troubleshooting Tree

Symptoms

Fuel Consumption Excessive.

How To Use This Tree

This symptom tree is to be used to troubleshoot fuel economy complaints. This tree is used along with the Fuel Consumption Customer Complaint Form and the Driveability Low Power/Excessive Fuel Consumption Checklist to help isolate engine, chassis, or driver issues associated with excessive fuel consumption.

Shoptalk

The Fuel Consumption General Information section of this manual and Troubleshooting Excessive Fuel Consumption, Bulletin 3666094, should be referenced prior to any troubleshooting being performed on a customer's engine.

The cause of excessive fuel consumption is difficult to diagnose and correct because of the potential number of factors involved. Actual fuel consumption problems can be caused by any of the following factors:

- Engine factors
- Vehicle factors and specifications
- Environmental factors
- Driver technique and operating practices
- Fuel system factors
- Low power or driveability problems.

Before troubleshooting, it is important to determine the exact complaint. Is the complaint based on whether the problem is real or perceived, or does **not** meet driver expectations? The Fuel Consumption Customer Complaint Form is a valuable list of questions that can be used to assist the service technician in determining the cause of the problem. Complete the form before troubleshooting the complaint. The following are some of the factors that **must** be considered when troubleshooting fuel consumption complaints.

- Excessive Idling Time: Idling the engine can use from 0.5 to 1.5 gallons per hour depending on the engine idle speed.
- Vehicle Aerodynamics: The largest single power requirement for a truck is the power needed to overcome air resistance. As a general rule, each 10 percent reduction in air resistance results in a 5 percent increase in mile per gallon.

- **Rolling Resistance:** Rolling resistance is the second largest consumer of power on a truck. The type of tire and tread design have a sizeable effect on fuel economy and performance. Changing from a bias ply to a low profile radial tire can reduce rolling resistance by about 36 percent.
- **Environmental and Seasonal Weather Changes:** There can be as much as 1 to 1.5 mile per gallon difference in fuel consumption, depending on the season and the weather conditions.
- **Truck Route and Terrain:** East and west routes experience almost continual crosswinds and head winds. Less fuel can be used on north and south routes where parts of the trip are not only warmer, but have less wind resistance.
- **Driver Technique and Operating Practices:** A 1 mile per hour increase in road speed equals a 0.1 mile per gallon increase in fuel consumption. This means that increasing road speed from 50 to 60 mph will result in a loss of fuel mileage of 1 mpg.
- **Result of a Low Power or Driveability Problem:** An operator will change driving style to compensate for a low power or driveability problem. Some things the driver is likely to do are (a) shift to a higher engine rate per minute (rpm) or (b) run on the droop curve in a lower gear instead of upshifting to drive at part-throttle conditions. These changes in driving style will increase the amount of fuel used.

Additional vehicle factors, vehicle specifications, and axle alignment can also affect fuel consumption. For additional information on troubleshooting fuel consumption complaints, refer to Troubleshooting Excessive Fuel Consumption, Bulletin 3387245.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Verify the complaint.	
	STEP 1A. Fill out the Fuel Consumption Customer Complaint Form.	N/A
STEP 2.	Use INSITE™ electronic service tool for electronic checks.	
	STEP 2A. Check for fault codes.	N/A
	STEP 2B. Confirm Features and Parameters.	N/A
	STEP 2C. Check the engine control module (ECM) calibration.	N/A

STEPS		SPECIFICATIONS
	STEP 2D. Monitor vehicle speed.	N/A
	STEP 2D-1. Inspect the engine and chassis grounds.	N/A
STEP 3.	Fuel system inspection.	
	STEP 3A. Inspect the vehicle for fuel leaks.	N/A
	STEP 3B. Check the fuel inlet restriction.	N/A
	STEP 3C. Check the drain line restriction.	N/A
	STEP 3D. Check the fuel grade.	N/A
	STEP 3E. Perform the INSITE™ Cylinder Performance Test.	N/A
	STEP 3E-1. Interpret the results of the Cylinder Performance Test.	N/A
	STEP 3E-2. Interpret the results of the Cylinder Performance Test.	N/A
	STEP 3E-3. Measure the resistance between the actuator and the actuator base.	N/A
	STEP 3E-4. Swap the front and rear metering actuator.	N/A
	STEP 3E-5. Swap the front and rear timing actuator.	N/A
STEP 4.	Check the air handling system.	
	STEP 4A. Check the air intake restriction.	N/A
	STEP 4B. Pressure test the charge-air cooler.	N/A
	STEP 4C. Inspect the charge-air cooler.	N/A
	STEP 4D. Check the exhaust restriction.	N/A

STEPS		SPECIFICATIONS
STEP 5.	Air handling diagnostic checks.	
	STEP 5A. Start the engine and read the fault codes.	N/A
	STEP 5B. Inspect the turbocharger blades.	N/A
	STEP 5C. Determination of turbocharger type.	N/A
	STEP 5D. Check the variable geometry actuator rod for correct travel.	N/A
	STEP 5D-1. Check for air leaks and inspect air lines.	N/A
	STEP 5D-2. Check the engine and vehicle grounds.	N/A
	STEP 5D-3. Check for air pressure at the turbocharger control valve outlet.	N/A
	STEP 5D-4. Check for air pressure at the turbocharger control valve outlet.	N/A
	STEP 5D-5. Check for correct turbocharger actuator travel.	N/A
	STEP 5D-6. Check for air pressure at the turbocharger control shutoff valve outlet.	N/A
	STEP 5D-7. Check for air pressure at the turbocharger control shutoff valve inlet.	N/A
	STEP 5D-8. Check for plugged turbocharger control shutoff valve filter.	N/A
	STEP 5E. Perform INSITE™ electronic service tool exhaust gas recirculation (EGR) Valve/Turbocharger Operational Test.	N/A
	STEP 5E-1. Check the engine and vehicle grounds.	N/A
	STEP 5F. Inspect the wastegate actuator hose.	N/A

STEPS		SPECIFICATIONS
	STEP 5G. Inspect the wastegate actuator rod for travel.	N/A
	STEP 5G-1. Inspect the wastegate actuator rod for travel.	N/A
	STEP 5H. Measure the resistance of the four-stage wastegate controllers, if equipped.	N/A
	STEP 5I. Inspect the four-stage wastegate controller, if equipped.	N/A
STEP 6.	Base engine checks.	
	STEP 6A. Verify the overhead adjustments are correct.	N/A
	STEP 6B. Verify the dipstick calibration.	N/A
	STEP 6C. Analyze the engine oil.	N/A
	STEP 6D. Check for internal engine damage.	N/A

STEP 1. Verify the complaint.

STEP 1A. Fill out the Fuel Consumption - Customer Complaint Form.

Conditions : N/A		
Action	Specification/Repair	Next Step
Fill out the Fuel Consumption - Customer Complaint Form in the Troubleshooting and Repair Manual, Signature, ISX, and QSX15 Engines, Bulletin 3666239. N/A	Problem caused by vehicle factors, environmental factors, or driver technique? YES	Repair complete
	Problem caused by vehicle factors, environmental factors, or driver technique? NO	2A

STEP 2. Use INSITE™ electronic service tool for electronic checks.

STEP 2A. Check for fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Use INSITE™ electronic service tool to read the fault codes. N/A	Any active or high counts of inactive fault codes? YES	Troubleshoot fault codes.
	Any active or high counts of inactive fault codes? NO	2B

STEP 2B. Confirm features and parameters.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Confirm the Programmable Features and Parameters are set correctly. Verify the following features and parameters: • Maximum Cruise Control Speed • Cruise Control Upper Droop • Cruise Control Lower Droop	Features and parameters set correctly? YES	2C
• Gear-Down Maximum Vehicle Speed (Heavy Engine Load) = (peak horsepower rpm x 60 min/hr) / (Gear Down Transmission Gear Ratio x Rear Axle Ratio x Tire Size) • Gear-Down Maximum Vehicle Speed (Light Engine Load) = (peak torque rpm x 60 min/hr) / (Top Gear Transmission Gear Ratio x Rear Axle Ratio x Tire Size) • Maximum Accelerator Vehicle Speed • Road Speed Governor Upper Droop • Road Speed Governor Lower Droop. Refer to the Feature Help Description in INSITE™ for additional information.	Features and parameters set correctly? NO Repair : Reset the Features and Parameters to the appropriate values.	Repair complete

STEP 2C. Check the ECM calibration.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Compare the ECM Code with the engine rating and CPL. Verify the calibration is correct. Use the calibration notes field in INSITE™ electronic service tool to confirm the calibration is correct for the application and control parts list (CPL). Verify the calibration is not set up for STx, LBSC, or Multiplexing if the application does not require it.	Calibration correct? YES	2D
	Calibration correct? NO Repair : Recalibrate the ECM with the correct ECM Code.	Repair complete

STEP 2D. Monitor vehicle speed.**Conditions :**

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Start the engine.

Action	Specification/Repair	Next Step
Monitor vehicle speed with INSITE™ electronic service tool while the vehicle is not moving. N/A	Vehicle speed read zero when the vehicle is not moving and the engine is operating? YES	3A
	Vehicle speed read zero when the vehicle is not moving and the engine is operating? NO	2D-1

STEP 2D-1. Inspect the engine and chassis grounds.

Conditions : • Turn keyswitch OFF.		
Action	Specification/Repair	Next Step
Check for loose or corroded engine, chassis, or battery ground connections. • Check the engine ground connection. • Check the chassis ground connections. • Check the battery terminal connections.	All grounds present, properly grounded, free of corrosion, and tight? YES Repair : Check the vehicle speed sensor (VSS) and the VSS circuit. Refer to Procedure 019-090 (/qs3/pubsys2/xml/en/procedures/99/99-019-090.html) and Refer to Procedure 019-091 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-091.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 3666259.	Repair complete
	All grounds present, properly grounded, free of corrosion, and tight? NO Repair : Replace, clean, or tighten the grounds.	Repair complete

STEP 3. Fuel system inspection.

STEP 3A. Inspect the vehicle for fuel leaks.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check the suction fuel lines, original equipment manufacturer (OEM) fuel connections, suction-side fuel filter, fuel pump, integrated fuel system module, and actuators for leaks. With EGR Check for pressure-side fuel filter, fittings and hoses to and from the ECM, cooling plate, fuel pump, integrated fuel system module, and actuators for leaks. Dye can be placed in the fuel to assist in finding fuel system leaks.	Any leaks present? YES Repair : Repair or replace the failed component.	Repair complete
	Any leaks present? NO	3B

STEP 3B. Check the fuel inlet restriction.

Conditions :

- Connect vacuum gauge to the suction-side Compucheck® fitting.
- Turn keyswitch ON.
- Operate engine at high idle.

Action	Specification/Repair	Next Step
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If the engine uses a fuel lift pump, wait until after priming pump has turned off and observe the reading on the vacuum gauge. N/A	Fuel inlet restriction less than the specifications listed below? • Dirty: 305 mm-Hg [12 in-Hg] • New: 203 mm-Hg [8 in-Hg]. YES	3C
	Fuel inlet restriction less than the specifications listed below? • Dirty: 305 mm-Hg [12 in-Hg] • New: 203 mm-Hg [8 in-Hg]. NO Repair : Locate the cause of high fuel inlet restriction. Check the suction-side fuel filter and fuel supply lines.	Repair complete

STEP 3C. Check the drain line restriction.

Conditions :

- Connect pressure gauge, Part Number 3375278.
- Turn keyswitch ON.
- Operate engine at high idle.

Action	Specification/Repair	Next Step
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Observe the reading on the pressure gauge. N/A	Fuel drain line restriction less than 229 mm-Hg [9 in-Hg]? YES	3D
	Fuel drain line restriction less than 229 mm-Hg [9 in-Hg]? NO Repair : Locate the cause of the high fuel drain line restriction in the OEM fuel return line.	Repair complete

STEP 3D. Check the fuel grade.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Verify the fuel grade used is appropriate for the application and ambient conditions and fuel quality is acceptable. Run the engine on an alternate, known good fuel source and verify engine performance if the fuel can not be tested appropriately. Refer to Fuel for Cummins Engines, Bulletin 3379001 (/qs3/pubsys2/xml/en/bulletin/3379001.html).	Fuel grade used and the quality acceptable? YES	3E
	Fuel grade used and the quality acceptable? NO Repair : Instruct the driver or owner to refuel with acceptable fuel.	Repair complete

STEP 3E. Perform the INSITE™ Cylinder Performance Test.

Conditions :

- Connect INSITE™ electronic service tool.
- Operate the engine at low idle.

Action	Specification/Repair	Next Step
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Perform the INSITE™ Cylinder Performance Test. Do not operate the engine above 900 rpm when performing the Cylinder Performance Test.	Do all injectors and both banks pass the Cylinder Performance Test? YES	4A
	Do all injectors and both banks pass the Cylinder Performance Test? NO	3E-1

STEP 3E-1. Interpret the results of the Cylinder Performance Test.

Conditions : N/A		
Action	Specification/Repair	Next Step
Interpret the results of the INSITE™ Cylinder Performance Test. - If the Cylinder Performance Test displays bar graphs for each cylinder's contribution, use the INSITE™ help file to interpret the results. - If the Cylinder Performance Test displays pass or fail for each cylinder, run the test at least three times to verify the results. Read the test results under the Results section of the test.	Cylinder Performance Test find an individual failed cylinder? YES Repair : Replace the failed injector. Refer to Procedure 006-026 in Section 6 (/qs3/pubsys2/xml/en/procedures/-006-026.html).	Repair complete
	Cylinder Performance Test find an individual failed cylinder? NO	3E-2

STEP 3E-2. Interpret the results of the Cylinder Performance Test.

Conditions : N/A		
Action	Specification/Repair	Next Step

Interpret the results of the INSITE™ Cylinder Performance Test. • If the Cylinder Performance Test displays bar graphs for each bank's contribution, use the INSITE™ help file to interpret the results. • If the Cylinder Performance Test displays pass or fail for each bank, perform the test at least three times to verify the results. Read the Test Results under the Results section of the test.	Cylinder Performance Test find an individual failed bank? YES	3E-3
	Cylinder Performance Test find an individual failed bank? NO	4A

STEP 3E-3. Measure the resistance between the actuator and the actuator base.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the timing and fueling actuators.

Action	Specification/Repair	Next Step
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Measure the resistance between the actuator and the actuator mounting base. N/A	Greater than 100k ohms? YES	3E-4
	Greater than 100k ohms? NO Repair : Replace the failed fueling or timing actuator. Refer to Procedure 019-110 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-110.html) or Refer to Procedure 019-111 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-111.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete

STEP 3E-4. Swap the front and rear fueling actuator.

Conditions : N/A		
Action	Specification/Repair	Next Step

Swap the front and rear fueling actuator to determine if the failed bank of cylinders follows the actuator. - Refer to Procedure 019-110 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-110.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334 for removal and installation. - Perform the INSITE™ Cylinder Performance Test.	Does the failed bank follow the metering actuator? YES Repair : Replace the failed fueling actuator. Refer to Procedure 019-110 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-110.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete
	Does the failed bank follow the metering actuator? NO	3E-5

STEP 3E-5. Swap the front and rear timing actuator.

Conditions : N/A		
Action	Specification/Repair	Next Step

Swap the front and rear timing actuators to determine if the failed bank of cylinders follows the timing actuator. - Refer to Procedure 019-111 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-111.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334 for removal and installation. - Perform the INSITE™ Cylinder Performance Test.	Does the failed bank follow the timing actuator? YES Repair : Replace the failed timing actuator. Refer to Procedure 019-111 in Section 19 (/qs3/pubsys2/xml/en/procedures/70/70-019-111.html), in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete
	Does the failed bank follow the timing actuator? NO Repair : Troubleshooting procedures need to be repeated from the beginning. A malfunctioning injector or actuator should have been detected.	3A

STEP 4. Check the air handling system.

STEP 4A. Check the air intake restriction.

Conditions :

- Run the engine at advertised horsepower and rpm.

Action	Specification/Repair	Next Step
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Check the intake system restriction by installing a vacuum gauge or water manometer into the air intake system. Refer to Procedure 010-031 in Section 10 (/qs3/pubsys2/xml/en/procedures/-010-031.html).	Air intake restriction greater than 635 mm-H₂O [25 in-H₂O] for a used filter or 254 mm-H₂O [10 in-H₂O] for a new filter? YES Repair : Correct the cause of high intake air restriction. Check for plugged air filter or restricted air intake piping.	Repair complete
	Air intake restriction greater than 635 mm-H₂O [25 in-H₂O] for a used filter or 254 mm-H₂O [10 in-H₂O] for a new filter? NO	4B

STEP 4B. Pressure test the charge-air cooler.

Conditions : Turn keyswitch OFF.		
Action	Specification/Repair	Next Step
Pressure test the charge-air cooler. Refer to Procedure 010-027 (/qs3/pubsys2/xml/en/procedures/-010-027.html) in Section 10.	Pressure drop 34 kPa [5 psi] or less in 15 seconds? YES	4C
	Pressure drop 34 kPa [5 psi] or less in 15 seconds? NO Repair : Repair the charge-air cooler.	Repair complete

STEP 4C. Inspect the charge-air cooler.

Conditions : Turn keyswitch OFF.
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Action	Specification/Repair	Next Step
Inspect the charge air cooler for bent or plugged fins. Bent or plugged fins will force the temperature of the air exiting the cooler to raise and force the engine to work harder to obtain the desired charge flow. A harder working engine will use more fuel.	Charge-air cooler free from bent or plugged fins? YES	4D
	Charge-air cooler free from bent or plugged fins? NO Repair : Clean the charge-air cooler.	Repair complete

STEP 4D. Check the exhaust restriction.

Conditions :

- Run engine at advertised horsepower and rpm.

Action	Specification/Repair	Next Step
Check the exhaust restriction by installing a pressure gauge into the exhaust system just past the turbocharger. Refer to Procedure 011-009 (/qs3/pubsys2/xml/en/procedures/-011-009.html).	Exhaust restriction less than 76 mm-Hg [3 in-Hg]? YES	5A
	Exhaust restriction less than 76 mm-Hg [3 in-Hg]? NO Repair : Repair the exhaust for source of high restriction.	Repair complete

STEP 5. Air handling diagnostic checks.

STEP 5A. Start the engine and read the fault codes.

Conditions :

- Connect INSITE™ electronic service tool.
- Operate engine at low idle.

Action	Specification/Repair	Next Step
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Check the fault codes with the engine running. Use INSITE™ to read the fault codes.	Fault Codes 2355 or 2356 active? YES	Go to appropriate Fault Code troubleshooting tree.
	Fault Codes 2355 or 2356 active? NO	5B

STEP 5B. Inspect the turbocharger blades.

Conditions :

- Turn engine OFF.
- Remove intake and exhaust connections for turbocharger.

Action	Specification/Repair	Next Step
Inspect the compressor and turbine fins for damage or wear. Measure the turbocharger wheel axial movement. Check the radial movement of the turbocharger wheel. Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).	Damage found on turbocharger fins or was axial or radial movement out of specification? YES Repair : Replace the turbocharger. Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).	Repair complete
	Damage found on turbocharger fins or was axial or radial movement out of specification? NO	5C

STEP 5C. Determination of turbocharger type.

Conditions :

N/A

Action	Specification/Repair	Next Step
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Determine if the turbocharger is a wastegated or variable geometry turbocharger. N/A	Is the turbocharger a variable geometry turbocharger? YES	5D
	Is the turbocharger a variable geometry turbocharger? NO	5F

STEP 5D. Check the variable geometry actuator rod for correct travel.

Conditions : • Turn engine OFF. • Turn keyswitch ON. • Connect INSITE™ electronic service tool. • Verify air tanks are fully charged.

Action	Specification/Repair	Next Step
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Perform the Turbocharger Actuator Test using the INSITE™ electronic service tool. • Adjust the delay timer in the Turbocharger Actuator Test so rod movement can be observed. • Select the Retract Actuator position with INSITE™. • Mark or scribe the VG™ actuator rod at the base of the actuator. • Select the Extend Actuator position with INSITE™. • Measure the rod travel by measuring the distance from the base of the turbocharger actuator to the scribe mark.	Turbocharger actuator rod extend between 10.2 and 11.8 mm [0.402 and 0.465 in]? YES	5E
The turbocharger actuator should move quickly and crisply. If the actuator rod movement is slow, there could be a problem with the air supply, a faulty ground connection on the engine or chassis, or mechanical problems with the variable geometry turbocharger assembly.	Turbocharger actuator rod extend between 10.2 and 11.8 mm [0.402 and 0.465 in]? NO	5D-1

STEP 5D-1. Check for air leaks and inspect air lines.

Conditions :

- Turn engine OFF.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Verify vehicle air tanks are fully charged.

Action	Specification/Repair	Next Step
Perform the Turbocharger Actuator Test using INSITE™ electronic service tool. Select the Extend Actuator position. Listen for air leaks at the following components: • Turbocharger control valve • Turbocharger control shutoff valve inlet connection • Turbocharger control shutoff valve outlet connection • Turbocharger control valve inlet connection • Turbocharger control valve outlet connection • Turbocharger actuator	Air leaks found in the system? YES Repair : Repair air leaks.	Repair complete
• All air lines, including the OEM supply line to the turbocharger control shutoff valve, turbocharger control shutoff valve to the turbocharger control valve, and turbocharger control valve to the turbocharger actuator.	Air leaks found in the system? NO	5D-2

STEP 5D-2. Check the engine and vehicle grounds.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for loose or corroded engine, chassis, or battery ground connections. • Check the engine ground connection. • Check the chassis ground connections. • Check the battery terminal connections.	All grounds present, properly grounded, free of corrosion, and tight? YES	5D-3
	All grounds present, properly grounded, free of corrosion, and tight? NO Repair : Replace, clean, or tighten the ground.	Repair complete

STEP 5D-3. Check for air pressure at the turbocharger control valve outlet.**Conditions :**

- Turn engine OFF.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Verify vehicle air tanks are fully charged.

Action	Specification/Repair	Next Step

Perform the Turbocharger Actuator Test using INSITE™ electronic service tool. Select the Retract Actuator position. Remove the air line connection at the outlet of the turbocharger control valve. Install an M12 Compucheck® fitting, Part Number 3824843, at the outlet of the turbocharger control valve. Install an air pressure gauge that is capable of reading at least 1034 kPa [150 psi] Perform the Turbocharger Actuator Test using the INSITE™ electronic service tool. Select the Extend Actuator position.	Vehicle air tank pressure present at the turbocharger control valve outlet? YES	5D-4
	Vehicle air tank pressure present at the turbocharger control valve outlet? NO	5D-6

STEP 5D-4. Check for air pressure at the turbocharger control valve outlet.

Conditions :

- Turn engine OFF.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Verify vehicle air tanks are fully charged.

Action	Specification/Repair	Next Step

Perform the Turbocharger Actuator Test using INSITE™ electronic service tool. Select the Retract Actuator position. Remove the air line connection at the outlet of the turbocharger control valve. Install an M12 Compucheck® fitting at the outlet of the turbocharger control valve. Install an air pressure gauge that is capable of reading at least 1034 kPa [150 psi].	Pressure gauge read more than 15 psi after 5 minutes? YES Repair : Replace the turbocharger control valve. Refer to Procedure 019-388 in Section 19 (/qs3/pubsys2/xml/en/procedures/70/70-019-388.html), in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete
	Pressure gauge read more than 15 psi after 5 minutes? NO	5D-5

STEP 5D-5. Check for correct turbocharger actuator travel.

Conditions :

- Turn engine OFF.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Verify vehicle air tanks are fully charged.

Action	Specification/Repair	Next Step

Remove the variable geometry actuator from the turbocharger. Refer to Procedure 010-113 (/qs3/pubsys2/xml/en/procedures/-010-113.html) in Section 10. Damage to the turbocharger may result if the proper removal procedure is not followed. Perform the Turbocharger Actuator Test using INSITE™ electronic service tool. Select the Extend Actuator position.	Turbocharger actuator rod travel at least 12 mm [0.472 in]? YES Repair : Replace the turbocharger assembly. Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).	Repair complete
	Turbocharger actuator rod travel at least 12 mm [0.472 in]? NO Repair : Replace the turbocharger actuator. Refer to Procedure 010-113 in Section 10 (/qs3/pubsys2/xml/en/procedures/-010-113.html).	Repair complete

STEP 5D-6. Check for air pressure at the turbocharger control shutoff valve outlet.

Conditions : • Turn engine OFF. • Turn keyswitch ON. • Verify vehicle air tanks are fully charged. • Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Use INSITE™ to start the Turbocharger Actuator Test. Loosen the air line connection at the turbocharger control shutoff valve outlet.	Can air be heard escaping from the turbocharger control shutoff valve outlet? YES Repair : Replace the turbocharger control valve. Refer to Procedure 019-388 in Section 19, (/qs3/pubsys2/xml/en/procedures/70/70-019-388.html) in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete
	Can air be heard escaping from the turbocharger control shutoff valve outlet? NO	5D-7

STEP 5D-7. Check for air pressure at the turbocharger control shutoff valve inlet.

Conditions :

- Turn engine OFF.
- Turn keyswitch ON.
- Verify vehicle air tanks are fully charged.

Action	Specification/Repair	Next Step
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Verify the OEM air supply line is connected to the correct port on the turbocharger control shutoff valve. Loosen the air line connection at the turbocharger control shutoff inlet.	Can air be heard escaping from the turbocharger control shutoff valve inlet? YES	5D-8
	Can air be heard escaping from the turbocharger control shutoff valve inlet? NO Repair : Repair the air supply from the OEM air tanks.	Repair complete

STEP 5D-8. Check for a plugged turbocharger control shutoff valve filter.

Conditions :

- Remove the turbocharger control shutoff valve filter.
- Turn keyswitch ON.
- Verify vehicle air tanks are fully charged.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Use INSITE™ to start the Turbocharger Actuator Test. Listen for an air leak at the turbocharger control shutoff valve filter head.	Air leak present at the turbocharger control shutoff valve filter head? YES Repair : Replace the turbocharger control shutoff valve filter. Refer to Procedure 010-114 in Section 10 (/qs3/pubsys2/xml/en/procedures/-010-114.html).	Repair complete
	Air leak present at the turbocharger control shutoff valve filter head? NO Repair : Replace the turbocharger control shutoff valve. Refer to Procedure 019-386 in Section 19 (/qs3/pubsys2/xml/en/procedures/70/70-019-386.html), in the Troubleshooting Manual, Signature Fuel System, Signature Engines, Bulletin 4021334.	Repair complete

STEP 5E. Perform INSITE™ electronic service tool exhaust gas recirculation (EGR) Valve/Turbocharger Operational Test.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.
- Operate engine at low idle.

Action	Specification/Repair	Next Step
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Select the EGR/VGT™ Operational Test using INSITE™ electronic service tool. Choose the Turbocharger Actuator option under Test Choices.	Turbocharger Operational Test pass? YES	6A
	Turbocharger Operational Test pass? NO	5E-1

STEP 5E-1. Check the engine and vehicle grounds.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for a loose or corroded engine, chassis, or battery ground connection. • Check the engine ground connection. • Check the chassis ground connections. • Check the battery terminal connections.	All grounds present, properly grounded, free of corrosion, and tight? YES Repair : Replace the variable geometry turbocharger. Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).	Repair complete
	All grounds present, properly grounded, free of corrosion, and tight? NO Repair : Replace, clean, or tighten the ground.	Repair complete

STEP 5F. Inspect the wastegate actuator hose.

Conditions :

- Turn engine OFF.
- Remove the turbocharger if the wastegate actuator is inaccessible.

Action	Specification/Repair	Next Step
Inspect the integral wastegate actuator hose for cracks or holes. N/A	Are holes or cracks found in the wastegate actuator hose? YES Repair : Replace the wastegate actuator hose.	Repair complete
	Are holes or cracks found in the wastegate actuator hose? NO	5G

STEP 5G. Inspect the wastegate actuator rod for travel.**Conditions :**

- Turn engine OFF.
- Remove the wastegate actuator hose from the wastegate actuator.

Action	Specification/Repair	Next Step
Apply a regulated air supply of 310 kPa [45 psi] to the actuator and check for actuator movement. N/A	Wastegate actuator rod move? YES	5H
	Wastegate actuator rod move? NO	5G-1

STEP 5G-1. Inspect the wastegate actuator rod for travel.**Conditions :**

- Turn engine OFF.
- Remove the e-clip from the wastegate pin and disconnect the actuator rod.

Action	Specification/Repair	Next Step
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Apply a regulated air supply of 310 kPa [45 psi] to the actuator and check for actuator movement. N/A	Wastegate actuator rod move? YES Repair : Move the wastegate lever on the turbocharger back and forth and check for smooth operation. Replace the turbocharger assembly if the wastegate is seized. Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html).	Repair complete
	Wastegate actuator rod move? NO Repair : Replace the wastegate actuator. Refer to Procedure 010-050 in Section 10 (/qs3/pubsys2/xml/en/procedures/-010-050.html).	Repair complete

STEP 5H. Measure the resistance of the four-stage wastegate controllers, if equipped.

Conditions :

- Turn engine OFF.
- Disconnect the ring terminals from the four-stage wastegate controllers, if equipped.

Action	Specification/Repair	Next Step

Measure the resistance from the wastegate controller post to engine block ground. The wastegate controller solenoids must be between 20°C and 26°C [68°F and 78°F] before using the resistance specifications listed.	Wastegate controller solenoid resistances between 7.0 and 8.0 ohms? YES	5I
	Wastegate controller solenoid resistances between 7.0 and 8.0 ohms? NO Repair : Replace the failed wastegate controller. Refer to Procedure 010-109 in Section 10 (/qs3/pubsys2/xml/en/procedures/-010-109.html).	Repair complete

STEP 5I. Inspect the four-stage wastegate controller, if equipped.

Conditions :

- Turn engine OFF.
- Remove the four-stage wastegate controllers.

Action	Specification/Repair	Next Step
Check the valve disc, valve seat, and actuator disc for dirt, metal, debris, bonding separation, corrosion, cracks, or wear. N/A	Damage or debris found on the valve disc, valve seat, or actuator disc? YES Repair : Replace or clean failed components.	Repair complete
	Damage or debris found on the valve disc, valve seat, or actuator disc? NO	6A

STEP 6. Base engine checks.

STEP 6A. Verify the overhead adjustments are correct.

Conditions :

- Turn keyswitch OFF.
- Remove valve cover.

Action	Specification/Repair	Next Step
Measure the overhead settings. Valve lash reuse limits: • Intake valve lash: 0.102 mm [0.004 in] to 0.483 mm [0.019 in] • Exhaust valve lash: 0.508 mm [0.020 in] to 0.813 mm [0.032 in].	Overhead settings within the reuse limits? YES	6B
	Overhead settings within the reuse limits? NO Repair : Adjust the overhead settings. Refer to Procedure 003-004 in Section 3 (/qs3/pubsys2/xml/en/procedures/-003-004.html).	Repair complete

STEP 6B. Verify the dipstick calibration.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
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Verify the dipstick calibration. Drain, remove, clean, and install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (/qs3/pubsys2/xml/en/procedures/-007-025.html). Check the calibration of the lubricating oil dipstick. Refer to Procedure 007-009 in Section 7 (/qs3/pubsys2/xml/en/procedures/-007-009.html).	Dipstick calibrated properly? YES	6C
	Dipstick calibrated properly? NO Repair : Calibrate the lubricating oil dipstick. Refer to Procedure 007-009 in Section 7 (/qs3/pubsys2/xml/en/procedures/-007-009.html).	Repair complete

STEP 6C. Analyze the engine oil.

Conditions : Turn keyswitch OFF.		
Action	Specification/Repair	Next Step
Take an engine oil sample. N/A	Engine oil analysis show any signs of wear metals or contaminants? YES	Repair complete
	Engine oil analysis show any signs of wear metals or contaminants? NO Repair : Locate and replace the worn component or remove the source of contamination and repair the engine as necessary.	6D

STEP 6D. Check for internal engine damage.

Conditions : Turn keyswitch OFF.		
Action	Specification/Repair	Next Step

Remove the oil filter. Refer to Procedure 007-013 (/qs3/pubsys2/xml/en/procedures/-007-013.html) in Section 7. Inspect the oil filter for foreign material. Refer to Procedure 007-002 in Section 7 (/qs3/pubsys2/xml/en/procedures/-007-002.html).	Foreign material found in the oil filter? YES Repair : Locate the source of foreign material and repair the engine as necessary.	Repair complete
	Foreign material found in the oil filter? NO	Return to the troubleshooting steps or contact a local Cummins Authorized Repair Location, if all steps have been completed and checked again.